Calibration and Pricing of Athena Autocalls: Heston Model Using FFT and Monte Carlo

Class: INTRO TO THE MATH OF FINANCE (Year 2025, Fall Semester)

Instructors name: Prof. Mikhail Smirnov

Team members

Ethan Chemla (EC4003), Ha Pham (HTP2105), Jesse Jacob Price (JJP2224), Ningyuan Tang (NT2504), Raphael Demare (RD3213), Vijeet Prasad (VP2580)

Rehearsal: Nov 30th 7:30 pm, Math Lounge (all members present)

Technology: Python (SciPy, NumPy, Pandas, Matplotlib)

Problem Statement

- Problem: “Pricing a path-dependent structure is hard because volatility is itself stochastic, and simple models like Black–Scholes or Local Vol cannot capture this behavior which is important for forward skews in path dependent options.”

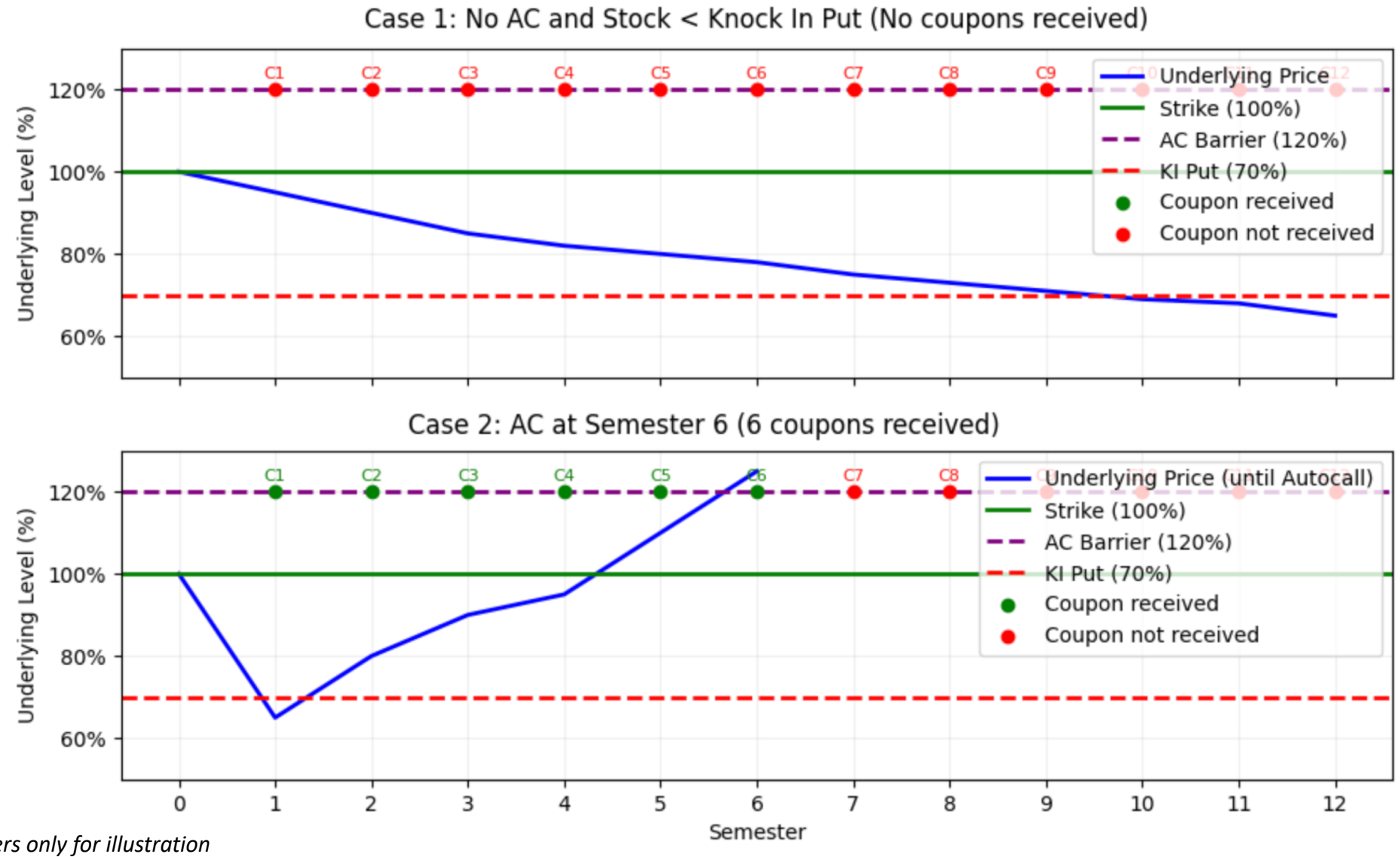
- Solution: “Use and calibrate a stochastic-volatility model (Heston) to match market vols and accurately price the path-dependent payoff.”

What Are Autocallables?

$$\text{Payoff} = \begin{cases} \text{ZCB} + \text{Cpn}, & S_{t_i} > B_{AC} \text{ for some } t_i, \\ \text{ZCB}, & B_{EKIP} < S_T \leq B_{AC}, \\ \text{ZCB} + N\left(\dfrac{S_T}{S_0} - 1\right), & S_T \leq B_{EKIP}. \end{cases}$$

- Periodic Autocall checks
- Pays step-up coupons on Autocall
- Downside risk at maturity
- Highly path-dependent → MC pricing required

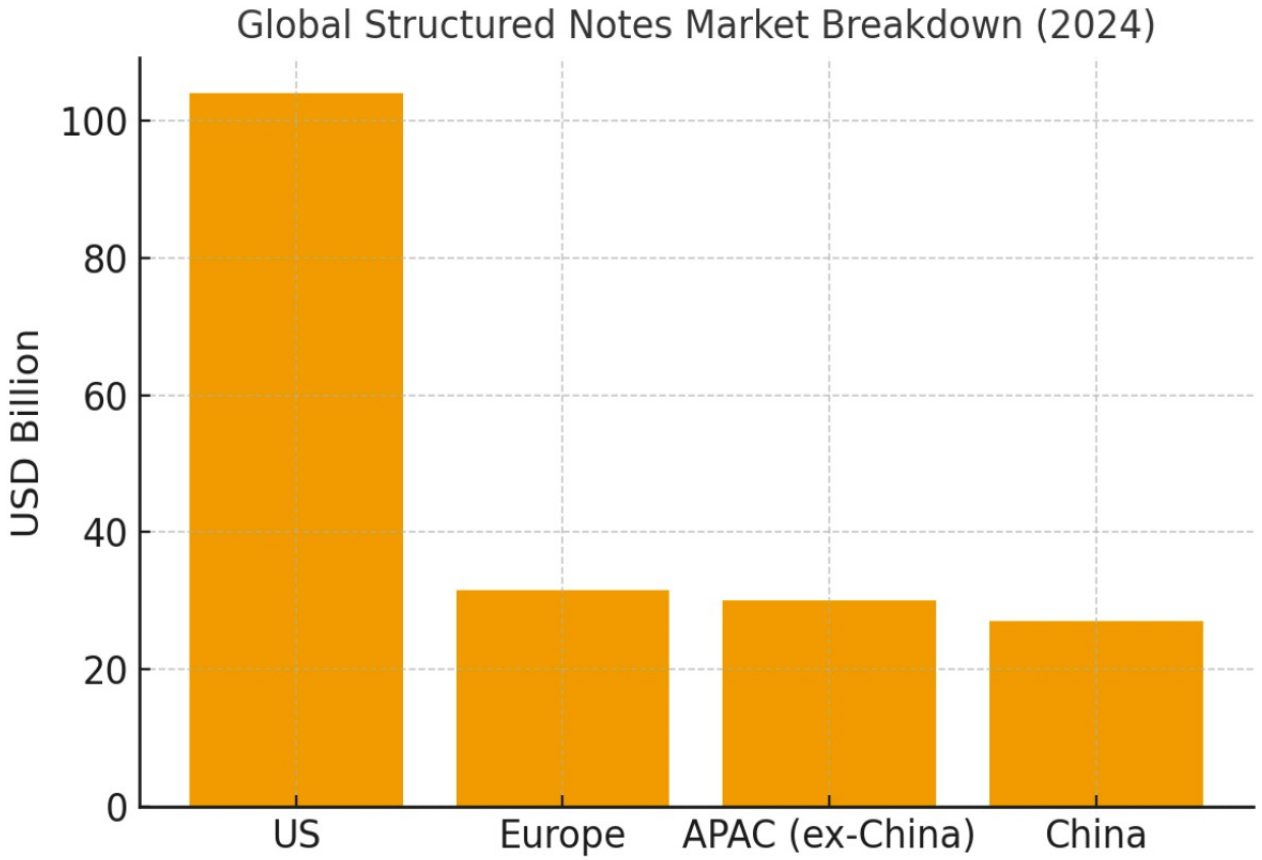
What Are Autocallables?



**Numbers only for illustration*

Why This Matters?

- US autocall issuance >$100B per year
- Autocallables ≈ 70% of structured-note issuance
- Global structured-notes market >$3T AUM
- SPX is one of the most common underlyings



Why the Heston Model?

- Formula

$$dS_t = rS_t\,dt + \sqrt{v_t}\,S_t\,dW_t^{(1)}$$

$$dv_t = \kappa(\theta - v_t)\,dt + \sigma_v\sqrt{v_t}\,dW_t^{(2)}$$

$$dW_t^{(1)}dW_t^{(2)} = \rho\,dt$$

- Heston captures:
  - Stochastic volatility
  - Mean reversion
  - Smile/skew
  - Term structure
  - Price–vol correlation

Term	Definition
S_t	underlying index level (SPX) at time t
ν_t	instantaneous variance at time t
r	risk-free interest rate
κ	speed of mean reversion of variance
θ	long-run mean variance
σ_v	volatility of variance (“vol-of-vol”)
ρ	correlation between price and variance shocks
$W_t^{(1)}, W_t^{(1)}$	Brownian motions under the risk-neutral measure

Calibration Objective

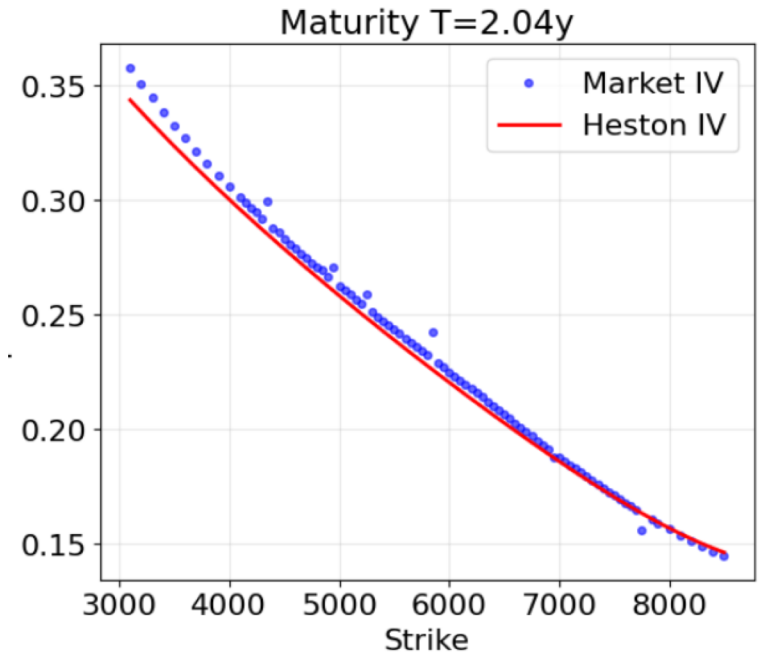
- Objective

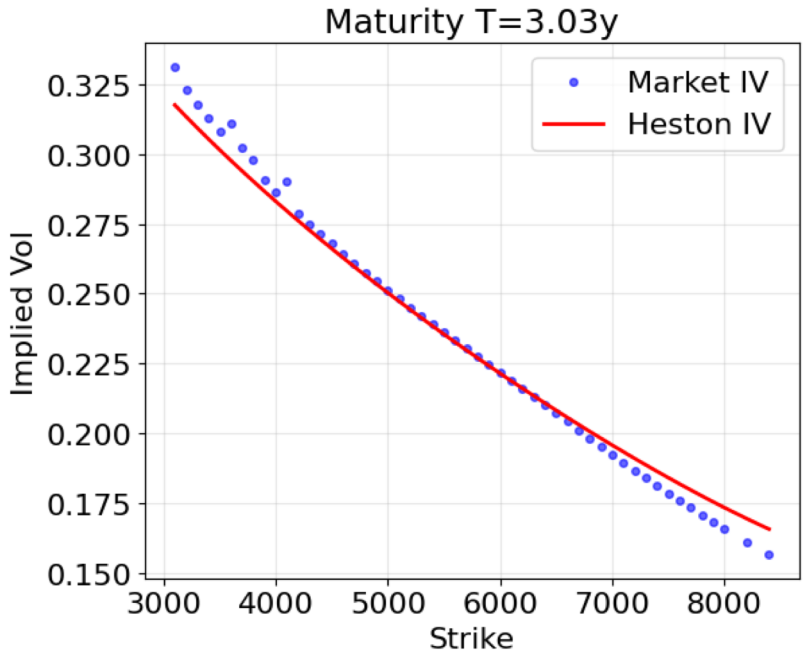
$$\text{RMSE: } \min_{\Theta=\{\kappa,\theta,\sigma,\rho,v_0\}} \left(\frac{1}{N} \sum_{i=1}^{N} (C_i^{\text{Market}} - C_i^{\text{Model}})^2 \right)^{1/2}$$

- Fit entire **strike–maturity grid**
- Weighted least squares
- Parameter vector: $(\kappa, \theta, \sigma_v, \rho, v_0)$
- Global search + local refinement for stability
- Calculation of C_i^{Model} using Fast Fourier Transform.

Iterate for each Maturity T: Frequency Domain [$\phi(u,T)$] $\longrightarrow$ **Damping & Discretization** $\longrightarrow$ **FFT Algorithm** $\longrightarrow$ **Price Domain** [Surface Row $C(K,T)$]

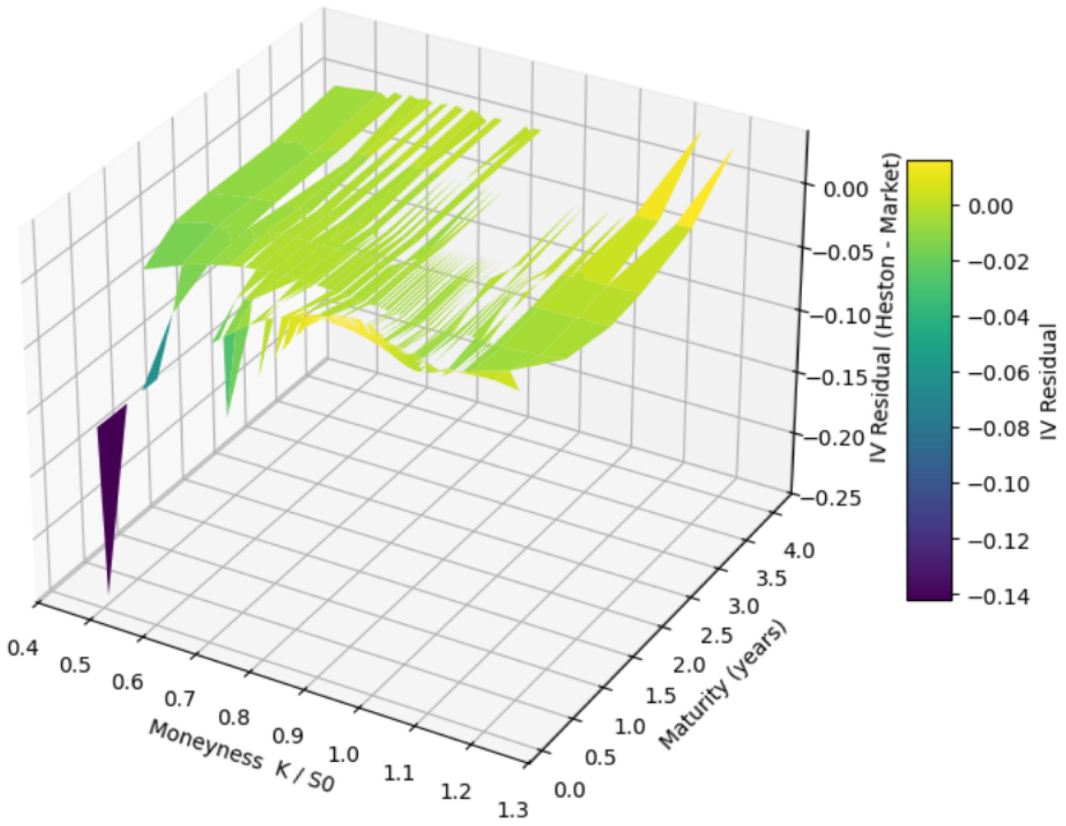
Calibration Results

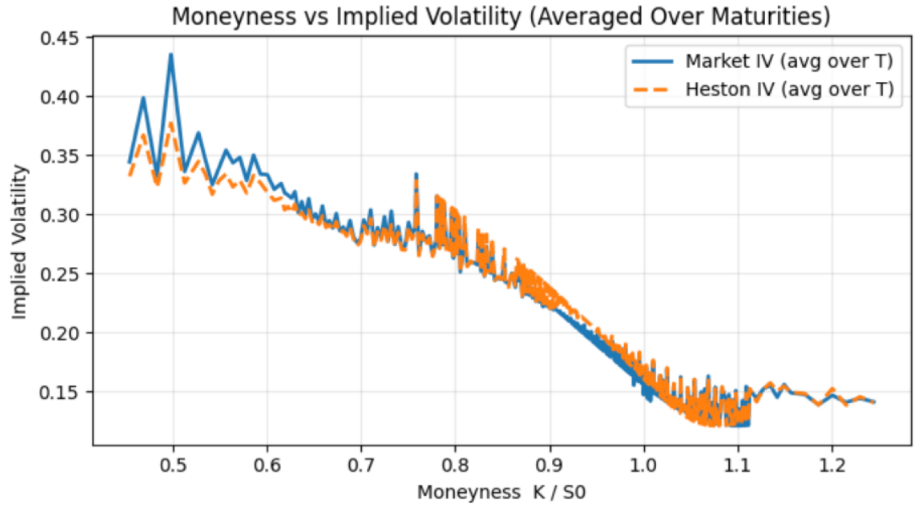


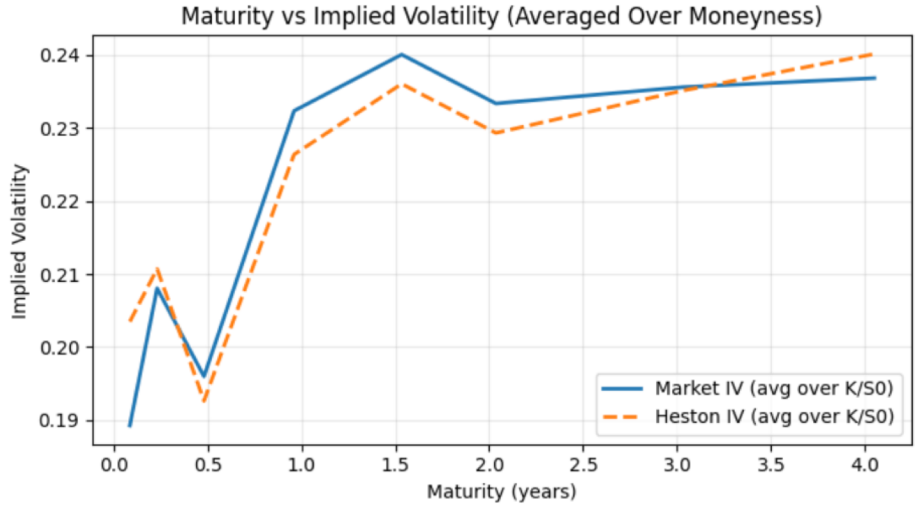


- Average RMSE across full surface: ~13
- Tight fit around ATM and mid-strikes
- Larger errors in deep ITM/OTM wings (expected for pure Heston)
- Term structure captured reasonably well

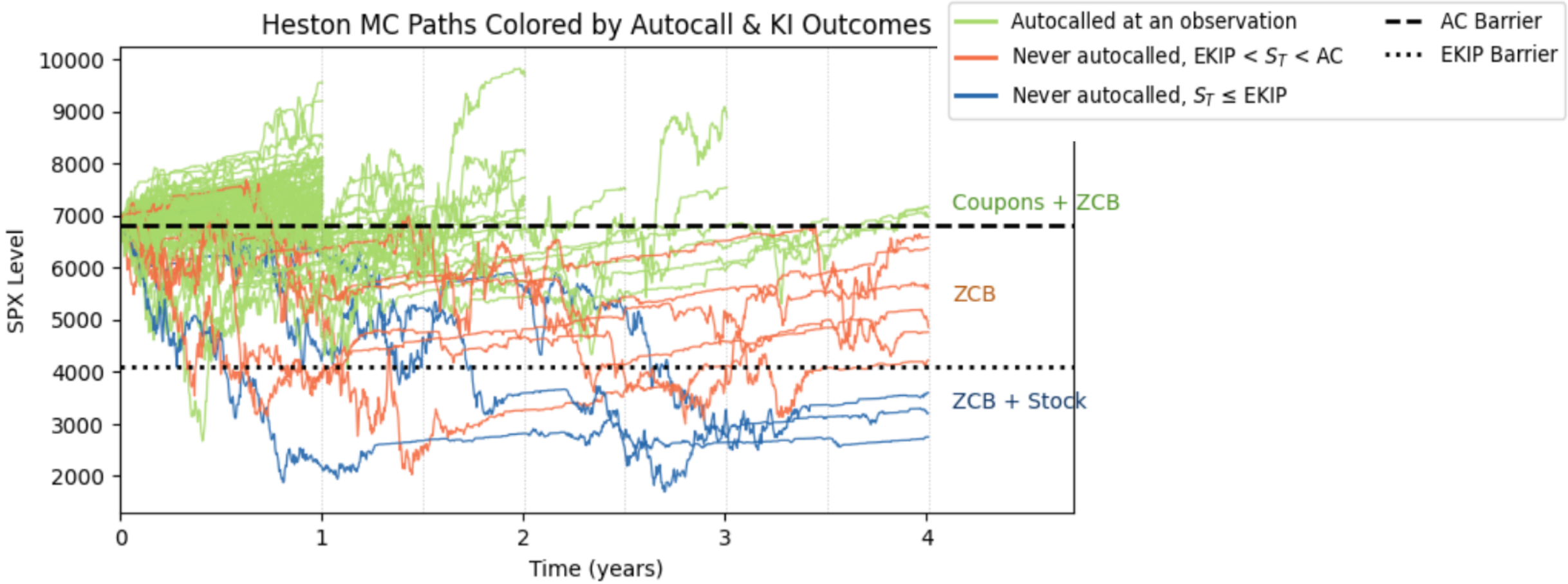
Calibration Diagnostics







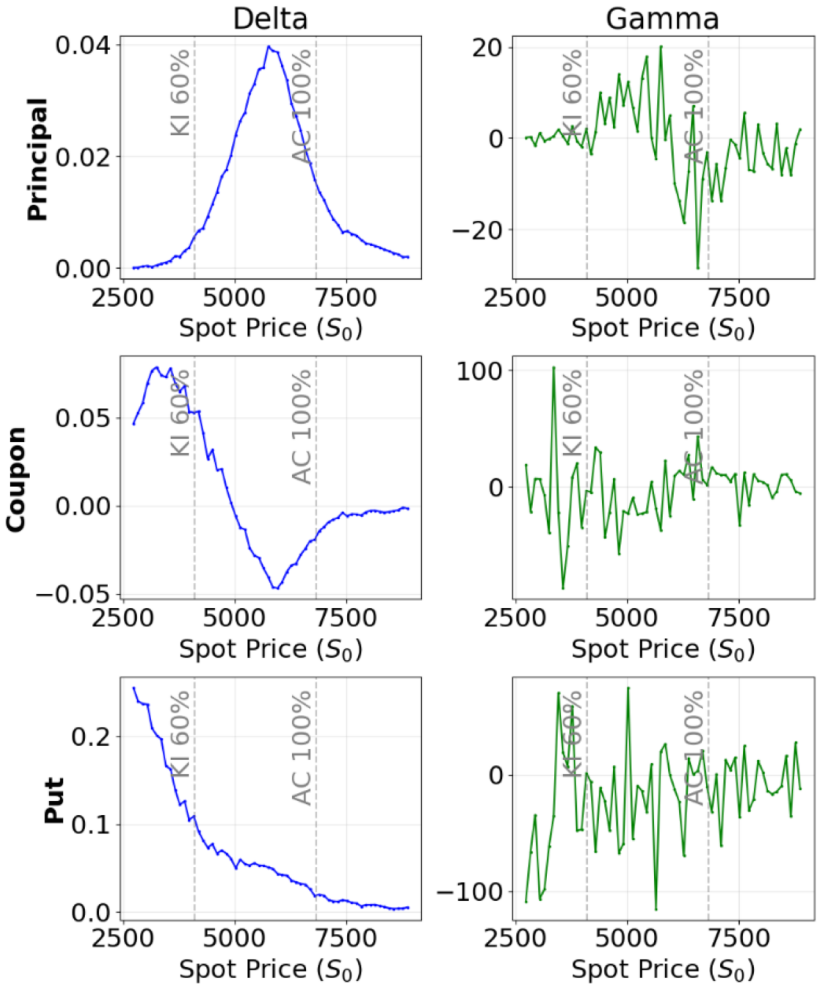
Pricing the SPX Athena Autocallable



$$\text{Price} \approx \mathbb{E}[\text{PV Payoff}] = \frac{N_{\text{AC}}\, e^{-rt_{\text{AC}}}(\text{ZCB}+\text{Coupons}) + N_{\text{Mid}}\, e^{-rT}(\text{ZCB}) + N_{\text{KI}}\, e^{-rT}(\text{ZCB}+\text{StockPerf})}{N_{\text{paths}}}$$

Pricing Results & Sensitivities

- Simulation complete: 50000 paths over 4 years (1011 steps).
- Nominal: $1 million
- Autocallable note price: $988.53k,
- Principal component: $947.81k,
- Coupon component: $73.79k,
- Downside Put Risk: -$33.07k
- Fair Value: $988.53k
- Percentage: 98.85%
- Expected life: 1.63 years



Conclusions & Extensions

- Conclusions
  - Heston calibration is stable; surface-wide RMSE ~13
  - Residual structure is smooth and concentrated in expected regions
  - Calibrated model supports realistic pricing of an SPX Athena autocallable
  - Workflow mirrors real structured-product desks

- Extensions
  - Local-stochastic volatility to improve wings
  - Pathwise/adjoint Greeks for hedging
  - Stress scenarios and regime analyses